(1.) Course Title: INFPHD458 Data Analytics Technologies and Methodologies applying to Enterprise Resource Planning Systems	**Credits: 6**
Course **Classification: Optional**	
Theoretical or Practical Nature of the Course, *'training character'*[13]: 3 theoretical – 3 practical credits (50-50t%)	
Type of Class: **lecture**: **18 hours** per **Semester** *(If the course is not (only) taught in Hungarian, then **the language**: **English**)* **Additional** (specific**) methods and characteristics**[2] used in delivering the knowledge *(if any)*: Case studies in SAP S/4HANA, SAP HANA 2.0, ABAP, SAP Signavio, SAP Build, Manufacturing Execution System; demonstrations using cloud-based analytics and data warehousing (SAC, Datasphere).	
Assessment Method (Exam / Midterm Grade / **Other**[3]): **continuous assesment** (in-semester evaluation; Additional assessment methods: individual and group assignments; case study presentation)	
Course Placement in Curriculum (Semester Number): 1st or 2nd semester	
Prerequisites (*if any*): (recommended: basic ERP and data management knowledge)	

Course Description: **A concise yet informative description of the knowledge** to be acquired.

The aim of the course is to provide comprehensive knowledge about data analytics, machine learning, natural language processing, large language models, and cloud-based analytics applied within enterprise resource planning systems. The course covers data representation, data quality, and modern data architectures with special emphasis on cloud-based data warehouses and their integration with ERP platforms.

Main modules / topics

- Data representation and data quality in ERP systems
- Cloud analytics and data warehousing architectures (HANA, Datasphere, SAC)
- Machine learning models and deep neural networks in ERP environments
- Large language models (LLM), NLP, and process mining with Signavio
- Integration of SAP technology platforms
- Enterprise case studies (S/4HANA, MES)
- Individual scientific / technological project
- Project presentation and evaluation

2-5 Most Important Required and Recommended Literature

List of required and recommended literature (notes, textbooks) with bibliographic data (author, title, publication details, (possibly pages), ISBN).

1. Suri, P. K. (2023). *AI-Powered Enterprise Resource Planning: A Holistic Approach for the Digital Age*. Boca Raton, FL: CRC Press.
2. Davenport, T. H. és Mittal, N. (2022). *All-in on AI: How Smart Companies Win Big with Artificial Intelligence*. Boston, MA: Harvard Business Review Press.
3. Linstedt, D. és Olschimke, M. (2015). *Building a Scalable Data Warehouse with Data Vault 2.0*. Waltham, MA: Morgan Kaufmann.

4. Fernández-Montero, J. A. (2023). *SAP HANA 2.0 Architectural Concepts: An Essential Guide for Technology Consultants* (2nd ed.). Bonn, Germany: SAP PRESS.
5. Ng, A. (2022). *Machine Learning Yearning: Technical Strategy for AI Engineers, In the Era of Deep Learning*. deeplearning.ai.

List of Prescribed Professional Competencies and Competency Elements

(knowledge, skills, etc., KKK Section 8) that the course significantly contributes to developing (highlight the most important ones).

a) Knowledge

- Has a research-level understanding of the general and specific characteristics, major directions, and precisely developed boundaries of the subject area within the discipline, as well as the agreed and debated interrelations.
- Creatively comprehends the interrelations, theories, and conceptual frameworks and terminologies of the given field of study.1
- Possesses research methodology knowledge necessary for independent research in the given discipline/field.

b) Skills

- Capable of creative analysis of the field, synthesizing comprehensive and specific interrelations with a new perspective, and conducting appropriate evaluative and critical activities.
- Able to apply and further develop the specialized learning and problem-solving methods of the field.
- Can creatively elaborate on the novel, previously unknown practical implications of theoretical issues.
- Capable of planning and executing new projects, conducting research in the field, and developing new techniques and approaches.
- Able to identify unforeseen professional problems and uncover the detailed theoretical and practical background necessary for their resolution at a research level.
- Can construct and communicate new relations relevant to the field, as well as comprehensive interrelations significant for personal and communal existence.

c) Attitudes

- Represents and further develops the relations that contribute to the process of human self-creation based on the field's characteristics.
- Possesses an interest and learning ability that enable identifying and solving currently unpredictable research problems in the field.
- Demonstrates a firm professional commitment, a continuous dedication to exploring new approaches, and an acceptance of the necessity for persistent effort.

d) Autonomy and Responsibility

[1] **Section 108**, Article 37 of the National Curriculum Act (Nftv.) 37. lesson: a session (lecture, seminar, exercise, consultation) requiring the personal involvement of the instructor to fulfil the academic requirements specified in the curriculum, the duration of which is at least forty-five minutes and not more than sixty minutes.
[2] e.g. case studies, role play, thematic presentations, etc.
[3] e.g. continuous debriefing, mid-year report
[4] e.g. request for case studies, thematic papers, essays, business, organisational plans, etc.

• Builds and initiates new knowledge areas with creative independence and proposes new practical solutions. • Participates in formulating theoretical and practical questions with a leadership role and high-level cooperation. • Engages as an equal discussion partner with professionals in the field. • Takes responsibility for raising and addressing new ethical questions concerning theoretical and practical aspects of the profession.
Course responsible (name, position, academic degree): **Dr. Gábor Tamás Orosz, Habil. Associate Professor, PhD**
Instructor(s) involved in teaching the course (if any) (name, position, academic degree): ……..….